Joshua C. Macdonald

Assistant Scientist · Johns Hopkins Bloomberg School of Public Health

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Research Profile

Computational scientist whose overarching interest is how ecological and landscape change influences the movement and connectivity of hosts, pathogens and cultural traits, and through them the persistence and evolution of structured populations. Established work addresses within- and between-host dynamics, including viral and immunological models from within-host to population scale, and cultural transmission; landscape-scale questions are a developing direction. Builds mechanistic models, inference methods and research software. Applications include infectious disease, public and global health and ecology; methods work also addresses incomplete data. Developing work includes scientific machine learning and the design of monitoring programs and studies with field and domain partners.

Research Themes

Within individuals: Viral growth, immune response and antibody kinetics and their consequences for transmission (foot-and-mouth disease, dengue; Crimean-Congo hemorrhagic fever in progress).

Across populations: Published work on cultural variation across geography and language. Developing: how landscape change, movement and contact shape spread, persistence and divergence.

Decisions: Surveillance and control decisions for livestock and wildlife disease, vaccination and scenario modeling under uncertainty, and study-design comparison.

Study Systems

Foot-and-mouth disease virus in African buffalo: Within-host model (first author) and carrier-transmission model (co-author); senior personnel on a multi-institution project.

Crimean-Congo hemorrhagic fever virus in livestock and wildlife: Collaborative transmission models and surveillance design.

Human cultural transmission: Networks and clusters of cultural traits across Austronesian-speaking societies; methods for incomplete cultural data.

Appointments

Johns Hopkins Bloomberg School of Public Health, Assistant Scientist 2026 – present
Baltimore, Maryland, USA
International Vaccine Access Center, Department of International Health, Global Disease Epidemiology and Control. Infectious disease modeling, inference, and research software development.

Johns Hopkins Bloomberg School of Public Health, Postdoctoral Scholar 2024 – 2026
Baltimore, Maryland, USA
International Vaccine Access Center, Department of International Health. Mentor: Shaun Truelove. Infectious disease modeling and inference for public-health applications.

Tel Aviv University, Zuckerman STEM Leadership Postdoctoral Fellow 2022 – 2024
Tel Aviv, Israel
School of Zoology. Mentor: Yoav Ram. Studied material culture and cultural transmission using Bayesian dimensionality reduction and multivariate analysis, drawing on archaeological training.

Education

PhD **University of Louisiana at Lafayette**, Mathematics 2017 – 2022
Lafayette, Louisiana, USA
- Advisor: Hayriye Gulbudak
- Dissertation: *Population Dynamics Under Disturbance in Ecology and Epidemiology*

MS **University of Louisiana at Lafayette**, Mathematics 2017 – 2020
Lafayette, Louisiana, USA

BA **University of North Carolina Asheville**, Mathematics — 2014 – 2016
Asheville, North Carolina, USA

BA **University of North Carolina Greensboro**, Archaeology — 2008 – 2013
Greensboro, North Carolina, USA

Published Papers

An Immuno-Epidemiological Model of Foot-and-Mouth Disease (FMD) in the African Buffalo Population with Carrier Transmission — 2026
S. Atkins, H. Gulbudak, J.S. Welker, H. Smith, ***J.C. Macdonald***, B.R. Beechler, A.E. Jolles
10.1007/s11538-026-01760-x (Bulletin of Mathematical Biology, 88, 194)

Cultural transmission, networks, and clusters among Austronesian-speaking peoples — 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
10.1017/ehs.2024.45 (Evolutionary Human Sciences, 6, e51)
- Software: VBPCApy · pp-eigentest project summary

Within-Host Viral Growth and Immune Response Rates Predict Foot-and-Mouth Disease Virus Transmission Dynamics for African Buffalo — 2024
J.C. Macdonald, H. Gulbudak, B. Beechler, E.E. Gorsich, S. Gubbins, E. Pérez-Martin, A.E. Jolles
10.1086/730703 (The American Naturalist, 204(2), 133–146)

Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms — 2023
J.C. Macdonald, H. Gulbudak
10.1007/s00285-023-01969-7 (Journal of Mathematical Biology, 87, 45)

Differential impacts of contact tracing and lockdowns on outbreak size in COVID-19 model applied to China — 2022
C.J. Browne, H. Gulbudak, ***J.C. Macdonald***
10.1016/j.jtbi.2021.110919 (Journal of Theoretical Biology, 532, 110919)

Modelling COVID-19 outbreaks in USA with distinct testing, lockdown speed and fatigue rates — 2021
J.C. Macdonald, C.J. Browne, H. Gulbudak
10.1098/rsos.210227 (Royal Society Open Science, 8, 210227)

Papers Under Review

Multi-Model Insights into Racial and Ethnic Heterogeneities in Early COVID-19 Transmission and Severity in the U.S. — 2026
S.J. Bents, S.L. Larsen, S. Jung, E. Howerton, et al., including ***J.C. Macdonald***
(American Journal of Epidemiology; revision submitted)

Preprints

Robust parameterization of a viral-immune kinetics model for sequential Dengue virus (DENV) infections with Antibody-Dependent Enhancement (ADE) — 2023
J.C. Macdonald, H. Gulbudak
10.1101/2023.08.29.555313 (bioRxiv)

Bayesian estimation of the number of significant principal components for cultural data — 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
10.48550/arXiv.2409.12129
- Software: pp-eigentest project summary · VBPCApy

Operational Modeling and Policy Support

Diphtheria Outbreak-Response Vaccination Modeling 2026 – present
Johns Hopkins IVAC. Lead modeler for multi-country, district-level transmission analyses comparing outbreak-response vaccination strategies across diverse diphtheria risk settings. Partner-specific policy processes and working materials remain private. Implemented with op_system, FlepiMoP2, JAX, and Diffrax.

COVID-19 Scenario Modeling Hub 2025 – present
ACCIDDA. Led the Round 20 (2026–27) submission, including rebuilding and migrating the legacy model to op_system and FlepiMoP2, model development and calibration, scenario production, quality assurance, and hub submission. Previously contributed data wrangling and data assimilation to the Hub's health-heterogeneities scientific round.
- COVID-19 Scenario Modeling Hub
- Submission repository

Flu Scenario Modeling Hub 2024 – present
ACCIDDA. Lead Model Developer across the 2024/25, 2025/26, and 2026/27 seasons. Set up the ACCIDDA flu model and operational pipeline (2024/25). Work with external modeling packages informed contributions to FlepiMoP2 (2025/26). Supervising an undergraduate student on revisions to the flu model for the 2026/27 round.
- Submission repository

Hib Vaccination Modeling 2026 – present
ACCIDDA. Technical supervisor for PhD students implementing an age- and immune-status-structured Haemophilus influenzae type b (Hib) model to evaluate long-running vaccination programs in a partner-defined population. Public descriptions remain general pending approval from participating organizations.

Research Software – Lead Developer

OP System (Operator-Partitioned System) 2026 – present
Turns readable model descriptions into calculations for simulation and data analysis. Handles spatial structure, interactions between groups and random events in disease, ecological and other dynamical models.
- Release status: Public; package maturity: alpha; latest released version: v0.10.0 (final). PyPI · Release archive · All versions
- Source

pp-eigentest 2026 – present
Research on distinguishing underlying patterns from noise in incomplete data. Revises a 2024 preprint and studies how missing observations and modeling assumptions affect that choice. Evaluation studies and manuscript revision are in progress.
- Release status: Private; package maturity: alpha; unreleased.
- Project summary

op_infer (Inference and Calibration) 2026 – present
Connects scientific models with data to estimate unknown quantities and compare data-collection plans. Tests fitting procedures on simulated data under stated modeling assumptions. Broader interfaces and application studies remain in development, and application materials remain private.
- Release status: Public; package maturity: pre-alpha; unreleased on PyPI (development version 0.0.1). Citation metadata
- Source · Project summary

Trade Study (Python) 2026 – present
Compares model settings, experiments and surveillance plans using repeated simulations. Helps researchers assess uncertainty and weigh competing goals such as accuracy, information and cost.
- Release status: Public; package maturity: alpha; latest released version: v0.6.0 (final). PyPI · Release archive · All versions
- Source

OP Engine (Operator-Partitioned Engine) 2025 – present
Simulates models that change over time and space, including systems with random events. Supports repeated simulations and summaries of quantities such as population counts. Capabilities and numerical checks depend on the selected method.
- Release status: Public; package maturity: alpha; latest released version: v0.9.1 (final). PyPI · Release archive · All versions
- Source

vbpca-py 2024 – present
Finds patterns in incomplete datasets without first filling in every missing observation. Produces low-dimensional summaries and variational covariance estimates, with tools for checking fit and choosing the number of summary dimensions.
- Release status: Public; package maturity: beta; latest released version: v0.5.0 (final). PyPI · Release archive · All versions
- Source

Research Software – Collaborative Infrastructure

FlepiMoP2 (Flexible Pipeline for Interchangeable Model Processing) 2025 – present
ACCIDDA. Coordinates simulation campaigns across locations and scenarios. Core contributor to its model, simulation and data-storage interfaces; led migration of ACCIDDA's COVID-19 model for the Round 20 Scenario Modeling Hub submission.
- Release status: Public; package maturity: alpha; latest released version: v0.3.0 (final). PyPI · Release archive · All versions
- Documentation

TabularEpi 2025
UNC / ACCIDDA. Tabular epidemic simulator supporting mixed deterministic-stochastic models. Conducted systematic performance benchmarking that identified bottlenecks, informing the design of the next-generation pipeline.

FlepiMoP (Flexible Epidemic Modeling Pipeline) 2024 – 2025
Hopkins IDD / ACCIDDA. Developed simulation and statistical inference components for operational influenza modeling. Performance profiling and vectorized backend work informed subsequent pipeline development.

Research Support – Active

USDA-NIFA AFRI US–UK Collaboration (Award #1032897) 2025 – 2028
Collaborator. Model parameterization, data analysis, and advice on mathematical and statistical methods. Quiescence to emergence: Does nairovirus diversity loss increase disease transmission, severity and the risk of emergence? US PIs: Brianna Beechler, Anna Jolles, Cameron Browne, and Hayriye Gulbudak; linked UKRI award BB/Z516934/1 with Emma Thomson, Teresa Lambe, Roman Biek, and Georgina Limon-Vega.

USDA-ARS Cooperative Agreement (3022-32000-021-018-S) 2025 – present
Collaborator. Models of transmission between species, surveillance design, and statistical advice. Investigating the Epidemiology of Crimean-Congo Hemorrhagic Fever Virus (CCHFV) in an Endemic Setting: Uganda. PIs: Bree Beechler and Anna Jolles (Oregon State University).
- USDA-ARS project record

CDC-RFA-FT-23-0069 (NU38FT000012) 2023 – present
Operational modeling, forecasting pipelines, and computational infrastructure. Atlantic Coast Center for Infectious Disease Dynamics and Analytics (ACCIDDA), an Insight Net center. Directors: Justin Lessler, Kimberly Powers, and Shaun Truelove.

NSF EEID / UKRI UK–US Collaboration Grant 2022 – 2027
Senior Personnel (from 2024). Model development and parameterization of within-host viral and immune dynamics. Multi-scale infection dynamics from cells to landscapes: FMD in African buffalo. Consortium-wide project budget: $4.05M over five years; this is not an amount directly controlled by me. PIs: Brianna Beechler, Roman Biek, Hayriye Gulbudak, Erin Gorsich, Simon Gubbins, Anna Jolles, and Jan Medlock.

Research Support – Completed

Zuckerman Postdoctoral Scholarship 2022 – 2024
Principal Investigator. Co-evolution of Cultural and Genetic Traits. Competitive postdoctoral fellowship; authored and defended the proposal. Total fellowship award: $120K.

Computational Expertise

Modeling: Mathematical and computational models of changing systems, including viral-immune, spatial, stochastic and compartmental models; machine learning with scientific constraints; multivariate data analysis and dimensionality reduction; stability, bifurcation and sensitivity analysis; parameter identifiability and inverse problems

Inference: Bayesian hierarchical models; variational and simulation-based inference; data assimilation; uncertainty quantification; simulation-based calibration, numerical qualification, and model and forecast assessment; proper scoring rules

Computing: Python, Julia, C++, Stan, R, MATLAB. Scientific computing; high-performance numerical solvers; operator splitting; data pipelines and exploratory analysis; reproducible workflows (Git, CI/CD, containerization)

Invited Talks and Seminars

Cross-Scale Feedback Motifs: Structure-Preserving Models and Computational Tools for Complex Systems Mar 2026
Biological Oceanography Seminar, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design Mar 2026
Differential Equations Seminar, University of Maryland Baltimore County, Baltimore, Maryland

Modeling long-term viral infections 2025
Mathematics Seminar, Shippensburg University, Shippensburg, Pennsylvania

Scenario and forecast modeling for respiratory viruses 2025
Navajo Nation Epidemiology Center Data Summit, Window Rock, Arizona

Impacts of host behavior and viral life history on infectious disease transmission 2024
Department of Mathematics, Virginia Tech; Infectious Disease Dynamics Group, Johns Hopkins University

Within-host viral life histories predict FMDV transmission dynamics for African buffalo 2024
Hawlena Lab, Ben-Gurion University, Midreshet Ben Gurion, Israel

Modeling process-based effects of harmful algal blooms 2023 – 2024
University of Essex (2024); Tel Aviv University (2023); Israel Oceanographic and Limnological Research, Haifa (2023); Inter-University Institute for Marine Sciences, Eilat (2023)

Patterns of cultural variation in the Austronesian expansion 2023
Fortunato Lab, University of Oxford (virtual)

Consilience in FMDV ecology 2022
Gresham Lab, New York University; Obolski Lab, Tel Aviv University

Selected Conference Presentations

Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design 2026
Society for Mathematical Biology Annual Meeting, Graz, Austria; mini-symposium talk

Recovering Ecological Geometry: A Trait- and Depth-Structured IPDE Model of Plankton Dynamics 2025
Society for Mathematical Biology Annual Meeting, Edmonton, Alberta, Canada; mini-symposium talk

Biologically Grounded Surrogate-Driven Parameter Inference for Sparsely Observed Dynamical Systems 2025
Symposium on Biomathematics and Ecology Education and Research (BEER), Fairfax, Virginia; contributed talk

Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms 2023
Society for Mathematical Biology Annual Meeting, Columbus, Ohio; mini-symposium talk

Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo 2022 – 2023
SIAM TXLA, Lafayette, Louisiana (2023); SMB Epidemiology-Immunology Midyear Meeting (2022, virtual)

Dengue immunodynamics: antibody-dependent enhancement, cross-reactive immunity, and model–data confrontation 2022 – 2023
SIAM TXLA (2023); Mathematical Perspectives on Immuno-Biology, Bulgaria (2023); Center for Combatting Pandemics, Tel Aviv (2023); European Conference on Theoretical and Mathematical Biology, Heidelberg (2022)

Disease dynamics necessarily operate across scales 2021
SIAM TXLA, South Padre Island, Texas; mini-symposium talk

Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates 2021
Society for Mathematical Biology Annual Meeting (virtual); contributed talk

Workshop Facilitation

Insight Net Third Annual Meeting, Tools Workshop Presenter Apr 2026
Chapel Hill, North Carolina, USA
Presented FlepiMoP2 and the op_system/op_engine simulation stack at the Insight Net Third Annual Meeting tools workshop, Friday Center, University of North Carolina at Chapel Hill.

Insight Net Second Tech Transfer Workshop, Technical Support Jan 2026
Chapel Hill, North Carolina, USA
ACCIDDA. Technical support for the Insight Net Second Tech Transfer Workshop at UNC Chapel Hill. Assisted participants with Python package installation and environment setup for Epydemics tools.

ACCIDDA, Workshop Facilitator Oct 2025
Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, October 15–17, 2025, Johns Hopkins Bloomberg School of Public Health.

ACCIDDA, Workshop Facilitator Aug 2024
Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, August 12–15, 2024, Johns Hopkins Bloomberg School of Public Health.

Teaching

Johns Hopkins University, Teaching Assistant 2025 – 2026
Baltimore, Maryland, USA
Teaching assistant for *Topics in Infectious Disease Epidemiology* with Shaun Truelove and Sara Loo, Summer 2025 and Summer 2026.

University of Louisiana at Lafayette, Teaching Assistant and Instructor of Record 2017 – 2020
Lafayette, Louisiana, USA
Instructor of record for elementary statistics (3 sections), survey of calculus (2 sections), precalculus (2 sections), and college algebra (2 sections). Teaching assistant for calculus, linear algebra, differential equations, and basic statistics.

McDowell County Schools / Memorial University / UNC Asheville, Teaching and Tutoring 2016 – 2017
Instructor of record for North Carolina Math III / Algebra II (2017). Mathematics and statistics tutoring at Memorial University of Newfoundland (2016–2017) and University of North Carolina Asheville (2016).

Mentoring

Mentoring students and postdoctoral researchers 2021 – present
- Jolles and Beechler Labs, Oregon State University (2025–present)
- Society for Mathematical Biology (2022–present)
- Association for Women in Mathematics (2023–2025)
- Ram Lab, Tel Aviv University (2022–2024)
- MIDAS Network (2021–2022)
- Gulbudak Group, University of Louisiana at Lafayette (2021–2022)

Undergraduate Research Supervision: Shany Naim 2024
Tel Aviv University. Supervised undergraduate research project: initial translation of the VBPCApy variational Bayesian PCA codebase from MATLAB to Python.

Community Outreach

Science communication and research dissemination 2021 – 2022
- University of Louisiana R1 Showcase: selected to present COVID-19 modeling research at the university-wide showcase (2022)
- Ocean View United Methodist Church, Oak Island, NC: community discussion on vaccines and COVID-19, with on-site pharmacist for vaccination (2021)

Professional Service

SMB Diversity, Equity, and Inclusion Committee 2023 – 2025
Member

Session Chair / Leader 2021 – 2023
- SMB Early Career Workshop: Applying for postdoctoral funding (2023)
- SMB Epidemiology-Immunology Midyear Meeting: Contributed talks (2022)
- SMB Annual Meeting: CT04-MEPI contributed talks in mathematical epidemiology (2021)

Mini-symposium Co-organizer 2019 – 2023
- SIAM TXLA: Data-driven mathematical and statistical modeling approaches to population biology (2023)
- SMB Annual Meeting: Disease Dynamics Across Scales (2023)
- SMB/ESTMB Joint Meeting: Multi-scale and data-driven modeling approaches in ecology, immunology, and epidemiology (2022)
- SIAM TXLA: Reproducibility, reliability, and robustness: confronting models from across mathematical biology with data (2021)
- SIAM TXLA: Population Dynamics in Ecology and Epidemiology (2019)

Secretary, University of Louisiana AMS Graduate Student Chapter 2021 – 2022

Journal Reviewing

2022–present: *PLOS ONE*, *Journal of Biological Systems*, *Journal of Biological Dynamics*, *Frontiers in Public Health*, *Frontiers in Veterinary Science*, *Computational and Applied Mathematics*, *Mathematical Biosciences*, *Scientific Reports*, *PLOS Computational Biology*, and *BMC Infectious Diseases*.

Honors and Awards

Travel Awards 2019 – 2022
SIAM, Society for Mathematical Biology, and University of Louisiana Graduate Research Association

Rhodes Outstanding Graduate Assistant Award 2021
Department of Mathematics, University of Louisiana at Lafayette. Awarded to two graduate students annually (approximately 50 graduate students total).

Outstanding Contributed Talk Award (Mathematical Epidemiology) 2021
Society for Mathematical Biology Annual Meeting

Three Minute Thesis Competition, Finalist 2019
Louisiana Sea Grant

Professional Affiliations

Society for Mathematical Biology; Society for Industrial and Applied Mathematics; International Society for Bayesian Analysis; United States Research Software Engineer Association; MIDAS Network; Association for Women in Mathematics; Institute of Industrial and Systems Engineers; Society for Health Systems (SHS/IISE); and Healthcare Information and Management Systems Society (HIMSS).